

Intelligent Evaluator – Project Summary

Project Type: AI-Assisted Code Evaluation

Domain: Programming Education

Technologies: Python (Flask), React, TypeScript, SQLAlchemy, JWT, LLMs, Streamlit

Overview

Intelligent Evaluator is a secure, scalable web platform designed to manage and automatically evaluate Python programming challenges in an academic environment. The system supports two roles—Professor and Student—using JWT-based authentication and role-based access control.

Key Features

- **Role-Based Workflow:** Professors create, publish, and manage challenges; students access published challenges and submit solutions.
- **Flexible Submissions:** Supports both code paste and .py file uploads with cross-platform encoding handling.
- **AI-Powered Evaluation:** Uses a semantic Large Language Model (LLM) to evaluate student code in the context of the problem statement, generating a score, concise feedback, and a summary.
- **Hybrid Architecture:** Designed to support optional runtime test hooks alongside semantic evaluation for extensibility.
- **Deep Evaluation Module:** One-click launch of a Streamlit-based analysis tool providing insights into code quality, complexity, maintainability, and security.

System Architecture

Frontend: React + TypeScript (Vite), Tailwind CSS, role-specific dashboards.

Backend: Flask REST APIs with SQLAlchemy ORM, JWT authentication, secure file handling.

Database: Structured models for users, challenges, and submissions.

Security: bcrypt password hashing, CORS protection, ORM-based SQL injection prevention, sandboxed code evaluation.

Impact & Learning Outcomes

- Improved scalability and fairness in programming assessment.
- Practical application of LLMs in real-world evaluation workflows.
- Experience in secure backend systems, full-stack development, and AI-assisted tooling.

Relevance

The project aligns with modern engineering challenges involving AI integration, cloud-ready architectures, security-first design, and user-focused platforms.